WEEK 4 • DAY 1

Machine Learning Foundations

Learn • Practise • Explain • Debug

Date	26-Oct-2026	**Duration**	5 hours
Difficulty	Intermediate	**Deliverable**	ML problem classification sheet

1. Today's Learning Goals

- Explain Machine Learning (ML) and how it differs from traditional rule-based programming.
- Distinguish supervised learning from unsupervised learning.
- Identify features (inputs) and labels (targets) in a dataset.
- Explain why data is split into training and test sets.
- Classify business problems by the ML approach they may need.
- Spot common problem-framing mistakes and improve the explanation.

2. Learn the Concepts in Simple Words

Machine Learning: A way to build systems that learn patterns from examples and use those patterns to predict outcomes or discover groups. Example: learn from past transactions to flag suspicious new transactions.

Traditional programming vs ML: Traditional programming often applies rules written by a developer: input + rules → output. In ML, an algorithm learns patterns from examples and uses them on new data.

Supervised learning: Training examples include a known target answer (label). Common tasks are classification (predict a category) and regression (predict a number).

Unsupervised learning: There is no supplied target label. The algorithm looks for structure or patterns, such as groups of customers with similar behaviour.

Features: Input columns used by a model. For a house-price model, features might include area, location, and number of bedrooms.

Label / target: The answer the model is trained to predict in supervised learning. For the house example, the label is the known sale price.

Training set: The portion of data used to train or fit the model.

Test set: A separate portion kept aside to estimate how the model performs on data it did not train on. Avoid repeatedly tuning decisions against the test set.

Important reminder: Not every problem needs ML. A clear rule, database query, or dashboard may be simpler, cheaper, and easier to explain.

3. Suggested 5-Hour Learning Plan

Time	Activity
Hour 1	Learn ML basics and compare rule-based programming with ML.

Hour 2	Study supervised vs unsupervised learning, classification vs regression.
Hour 3	Practise identifying features, labels, training data, and test data.
Hour 4	Complete the 10 business-problem classification exercise.
Hour 5	Debug common mistakes, explain answers in your own words, and review submission.

4. Quick Practice: Features and Labels

Example: predicting whether a customer will renew a subscription.

Possible features: months as a customer, number of logins, support tickets, plan type.

Label: renewed (Yes/No), if historical renewal outcomes are available.

Patterns: we can look for patterns between customer activity and subscription renewal.

For each scenario below, identify possible features and the label (if one exists). If no label is supplied, explain what patterns or groups you might look for.

5. Main Hands-On Activity: Classify 10 Business Problems

For each scenario, choose a starting approach: supervised classification, supervised regression, unsupervised clustering, or not necessarily ML / rule-based analysis. Explain your choice in 1–3 sentences. Real choices depend on the business goal and available data.

1. Fraud flag — A payments company has historical transactions marked fraudulent or legitimate and wants to flag new transactions for review.

2. Sales forecast — A retailer wants to estimate next month's sales amount using historical sales and relevant business data.

3. Customer groups — A business has customer purchase and engagement data but no predefined categories. It wants to discover groups with similar behaviour.

4. Subscription renewal — An app has historical records showing whether each customer renewed and wants to predict renewal for current customers.

5. House price — A property platform has past sale records and wants to estimate a property's likely sale price.

6. Duplicate invoice check — A company wants to flag an invoice if its invoice number already exists for the same vendor in its system.

7. Product recommendations — An online store wants to discover products often viewed or purchased together, without a manually supplied target label.

8. Support ticket priority — A support team has past tickets labelled urgent or non-urgent and wants to predict the priority of new tickets.

9. Inventory demand — A shop wants to predict the number of units of a product likely to be sold next week.

10. Spending patterns — A financial app wants to group users by spending behaviour but has no existing segment labels.

Classification Worksheet

#	Chosen approach	Possible features / data	Label (if any)	Reason / assumptions
1 **.Fraud flag**	supervised learning	Amount ,time ,location and payment method	Fraudulent/legitimate	Historical transactions have labels, so the model can learn to classify new transactions.
2. **Sales forecast**	Supervised regression	Past sales, month, product demand, promotions	Next month's sales amount	The goal is to predict a numerical value, so regression is suitable.
3**. Customer groups**	Unsupervised clustering	Purchase history, spending, website activity	No labels	Clustering can group customers with similar behaviours without existing categories.
4**. Subscription renewal**	Supervised classification	Login frequency, subscription plan, customer duration	Renewed / Not renewed	Historical renewal labels can help predict whether current customers will renew.
5 **.House price**	Supervised regression	House location, size, number of rooms, house age	Sale price	The goal is to estimate a numerical price, so regression is suitable
6**. Duplicate invoice check**	Not necessarily ML / Rule-based analysis	Invoice number, vendor name, invoice records	Duplicate / Not duplicate (rule-based result)	The system can check whether the same invoice number already exists for the same vendor using a simple rule.
7**. Product recommendations**	Unsupervised pattern discovery	Products viewed, products purchased together, purchase history	No predefined label	The business can discover which products are frequently viewed or purchased together.

8**. Support ticket priority**	Supervised classification	Ticket text, issue type, customer details, previous tickets	Urgent / Non-urgent	Historical priority labels can train a model to classify new support tickets.
9**. inventory demand**	Supervised regression	Past sales, product type, season, promotions	Number of units sold next week	The goal is to predict a numerical quantity, so regression is suitable
10 **Spending patterns**	unSupervised regression	Monthly spending, transaction frequency, spending categories	No labels	Clustering can group users according to similar spending behaviours.

6. Training vs Test Data: Mini Exercise

Imagine you have 1,000 historical customer records with known renewal outcomes.

1. Suggest a reasonable train/test split and explain your choice. An 80/20 split is a common starting point, not a universal rule.
   Answer: I would split the 1,000 customer records into 80% training data and 20% testing data. So, 800 records are used for training and 200 records are used for testing. This is a common starting point.
2. Explain what the training set is used for and what the test set is used for.
   Answer: The training set is used to teach the model using historical data. The test set is used to evaluate how well the trained model predicts outcomes for data it has not seen before.
3. Name one way information could accidentally leak from the test set into training or model decisions.
   Answer: Data leakage can happen if test data is used while training the model or selecting its settings. This can make the model's test performance look better than it really is.
4. If records are time-dependent, explain why a time-based split may be more appropriate than a random split.
   Answer: Data leakage can happen if test data is used while training the model or selecting its settings. This can make the model's test performance look better than it really is.

7. Debugging: Identibfy and Improve the Mistakes

Mistake A: A student says: “We want to predict next month's sales amount, so this is classification.”

Your task: Correct the problem type and explain why.

Answer: Predicting next month's sales amount is a supervised regression problem, not classification. Regression is used because sales amount is a numerical value.

Mistake B: A dataset has age, monthly spend, and city. A student says age is the label simply because it is the first column.

Your task: Explain how a target is chosen and what information is missing.

Answer: A target or label is chosen based on what we want to predict. Age is not automatically the label just because it is the first column. We need to know the business goal, such as predicting monthly spending or customer category.

Mistake C: A student trains a model on all records and reports its score on those same records as proof it works well on new customers.

Your task: Explain why this is misleading and how a test set helps.

Answer: Evaluating a model on the same records used for training can give a misleadingly high score. We should use a separate test set to check how well the model performs on unseen data.

Mistake D: A team uses clustering but claims it can directly predict customer default even though no default labels were supplied.

Your task: Explain the difference between discovering groups and predicting a known target.

Answer: Clustering discovers groups of similar customers without predefined labels. It cannot directly learn to predict customer default from those groups alone. To train a supervised model to predict default, we generally need historical data labelled as default or no default.

8. Explain in Your Own Words

5. What is ML? Explain it to a friend who has never coded.
6. What is the difference between supervised and unsupervised learning?
7. What is the difference between a feature and a label? Give an example.
8. Why should we test a model on data it did not train on?
9. Give one business problem where simple rules are better than ML, and explain why.

Answer: __: Machine Learning (ML) is a branch of Artificial Intelligence that allows computers to learn

patterns from data and make predictions without being programmed for every situation._

Answer: : Supervised learning uses labeled data with correct answers, while unsupervised learning uses

unlabeled data to find patterns or groups._

Answer: Feature: An input detail used by a machine learning model to make a prediction.

Label: The target answer that the model learns to predict.

Example: To predict a student's result: - Features: Study hours and attendance. - Label: Pass

Answer: __We test a model on data it did not train on to check how accurately it can make predictions on

new, unseen

Answer: A shop can give a 10% discount to customers who purchase items worth more than ₹5,000.

Simple rules are better because they are easy to implement and do not require training data or an ML model._

9. Final Deliverable and Submission Rules

Submit ONE file: Week4_Day1_ML_Problem_Classification_YourName.docx or .xlsx

- Complete all 10 rows of the classification worksheet.
- Include possible features/data and label (if any) for each scenario.
- Answer the training/test mini exercise.
- Complete all four debugging questions.
- Write explanations in your own words and state any assumptions.
- Be ready to explain two of your classifications to the trainer.

10. Self-Check Before Submission

- I can explain ML vs rule-based programming.
- I can distinguish classification, regression, and clustering.
- I can identify possible features and labels.
- I understand why training and test data should be separated.
- I classified all 10 scenarios and explained my reasoning.
- I corrected the debugging mistakes and can explain my answers.

Assessment Rubric (100 Marks)

Assessment area	**Marks**
10 problems: approach + reasoning	40
Features, labels, and assumptions	20
Training/test mini exercise	10
Debugging and improvement	20
Own-words explanations and clarity	10
TOTAL	**100**

Trainer note: Accept well-reasoned alternatives when the student states sensible assumptions. Recommendation and fraud problems can be approached in different ways depending on labels, data, and business goals.